Expert and LLM Evaluation of Alzheimer's Literature RAG: Response-Level Agreement and System Rankings

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Abstract

Can an LLM judge that ranks RAG systems similarly to experts also evaluate their individual responses consistently with those experts? We examine this distinction in an Alzheimer's disease literature-RAG archive containing 50 researcher-authored queries, eight retrieval configurations, and ratings from five biomedical researchers and GPT-4o. We compare scores for factuality, diversity, clarity, and insightfulness at two levels: individual query–condition pairs and configuration averages. Across the eight configurations, Spearman rank correlations between expert and GPT-4o means were 0.67–0.93; across 400 query–condition pairs, correlations were 0.27–0.52. Expert-minus-GPT score offsets were positive for factuality (+0.63) and diversity (+0.13), and negative for clarity and insightfulness (both -0.52). Complete-case and rater-omission analyses retained positive response-level associations, although their magnitudes varied with panel composition. Single-rater intraclass correlations were 0.09–0.24. The comparison is exploratory and concerns two realised protocols: experts reviewed retrieved abstracts and conditions side by side, while GPT-4o scored individual query–response pairs without abstracts. The findings distinguish agreement in system rankings from agreement on individual answers and motivate reporting both, alongside score offsets and rater reliability. We release text-free paired ratings and scripts reproducing the analyses.

1 Background

Retrieval-augmented generation (RAG) conditions language-model responses on external documents and offers a practical way to support biomedical literature question answering [1–3]. Domain terminology and compositional information needs make retrieval challenging. Query rewriting and hypothetical document generation are established ways to modify retrieval inputs [4, 5]; biomedical systems also explore generative query expansion and iterative follow-up questions [6, 7]. These techniques motivate our archived Alzheimer's disease literature-RAG experiment, but response quality must be evaluated separately from retrieval effectiveness.

LLM judges make response evaluation scalable, yet their agreement with people depends on the task and protocol. Prior work on MT-Bench and Chatbot Arena examined agreement with human preferences and biases in automated judging [8]; subsequent work surveys reliability, consistency, and bias across evaluation settings [9]. Agreement observed in general conversational benchmarks cannot establish agreement with biomedical researchers assessing specialised literature responses. Moreover, a correlation between two sets of scores measures association, not equality of scores or factual correctness.

Medical-text evaluation studies already compare LLM judges with expert assessments. Chen et al. [10] examined this comparison for medical summarisation, including disagreement among

Shared query pool: 201 researcher-authored queries; 50 with expert ratings.

Eight configurations: BM25 or DPR, each with naive retrieval, HyDE, keyword HyDE, or MeSH-guided keyword HyDE.

Evidence and generation: two retrieval steps per stream; two document slots without keywords, usually four with keywords; GPT-4o-mini response generation.

Expert protocol: five researchers; query, response, and abstracts; conditions shown side by side.

GPT-4o protocol: query and response; no abstracts; one response at a time.

Analysis: 400 query–condition pairs; four scoring dimensions; query-cluster bootstrap and expert inter-rater reliability.

Figure 1: Study design and the two realised scoring protocols. Retrieval uses BM25 or DPR separately. MeSH is prompt guidance, and the two-step procedure does not implement multi-hop reasoning. Evidence access and presentation format differ between evaluation protocols.

experts. A healthcare scoping review found that reported human alignment varies across tasks and validation designs [11]. A practical question for literature-RAG evaluation is whether agreement in the average ranking of configurations also describes agreement on individual responses. The two comparisons serve different uses: selecting a configuration and assessing a particular answer.

Our archive allows both comparisons on the same responses. Five biomedical researchers evaluated eight RAG configurations for 50 Alzheimer's disease research queries, and GPT-4o supplied ratings on the same four dimensions. We ask three questions: how does score association differ between query–condition pairs and configuration averages; which dimensions show directional score offsets; and how sensitive are the response-level associations to missing scores and panel composition? Expert inter-rater reliability provides context for interpreting the panel mean.

The study contributes a paired, evaluator-level rating resource and a reproducible comparison of these distinct aspects of evaluation. Its main empirical finding is that relatively similar configuration rankings coexist with weaker response-level associations and dimension-specific offsets. Query reformulation supplies the eight configurations for this case study; development of a new retrieval algorithm is outside its scope. The analysis compares the realised expert and automated protocols, whose evidence access and presentation differed.

2 Methods

We retrospectively analysed archived responses and ratings from a query-reformulation experiment. The primary estimands are associations and mean score differences between expert-panel averages and GPT-4o ratings on the same query–condition pairs. Configuration scores provide context for this comparison; the design does not isolate causal effects of retrieval or reformulation choices.

For each of BM25 and DPR, the shared archive includes four configurations: the original query without keywords (naive), HyDE without keywords, key-concept keyword prompting with HyDE, and MeSH-guided keyword prompting with HyDE. Rewriting conditions are described for archive provenance but excluded from the paired analysis because expert ratings are unavailable.

2.1 Keyword Generation

The pipeline generated keywords from each query using GPT-4o-mini. The key-concept prompt requested specific biological concepts and excluded generic terms such as "relationship" and "mechanism". The MeSH-guided prompt instead asked for terminology consistent with Medical

Subject Headings, with an emphasis on genes, diseases, and biochemical pathways. This was prompt guidance: the implementation did not look up, validate, or expand terms against the MeSH ontology. The prompts are reproduced in Appendix A.1.

The prompt requested up to two keywords per query. Inspection of the recovered 201-query archive showed that 199 queries produced two keywords, while two queries produced five; the implementation retained all returned keywords rather than truncating them. Both exception queries were included in the 50-query human-evaluation subset. We therefore describe two keywords as the intended output, not an invariant of the realised experiment.

2.2 Query Transformation

The archived pipeline supported query rewriting [4] and hypothetical document generation [5]. Rewriting requested a reformulation of the original query, optionally conditioned on a generated keyword. These conditions are documented in the appendix but are excluded from the paired analysis because expert ratings are unavailable.

For HyDE, GPT-4o-mini generated a pseudo-document from the original query, either alone or with one generated keyword. The resulting text, denoted q^*, was used as the retrieval input. Keyword-conditioned runs created one retrieval stream per returned keyword. For naive conditions, q^* was the unmodified query. We use the archived label "HyDE" for pseudo-document conditions under both retrievers; under BM25, the pseudo-document was used as a lexical query rather than embedded as in dense HyDE. Retrieval relevance was not independently measured.

2.3 Two-Step Context-Augmented Retrieval

Each retrieval stream used two steps, shown in Equation (1), where d_1 and d_2 denote the selected documents.

$$\begin{aligned} \text{Retriever}(q^*) &\rightarrow d_1 \\ \text{Retriever}(\{q^*, d_1\}) &\rightarrow d_2 \end{aligned} \tag{1}$$

For each transformed-query stream, the intended procedure retrieved the top-ranked document d_1 for q^* ($k = 1$). It then concatenated q^* and d_1 to form an expanded context, which was used to retrieve a second document d_2. In keyword-conditioned implementations, this procedure is run independently for every returned keyword. No-keyword conditions therefore supplied two document slots to the generator, whereas keyword conditions usually supplied four slots. The two five-keyword exceptions supplied ten slots. Exact duplicates could remain across keyword streams.

Within a stream, the intended rule was to select the next-ranked result when the second-step top result duplicated d_1. A code audit identified an exception in the recovered BM25 no-keyword implementation: the second-ranked candidate was selected in both branches of the conditional, including when the top-ranked candidate was not a duplicate. We analyse the archived outputs as generated and do not retroactively correct this implementation artefact. No sub-question decomposition, evidence chaining, or multi-hop reasoning is performed.

2.4 Generation

GPT-4o-mini generated a response from the original query and the retrieved documents. Keyword-conditioned runs also passed the generated keywords directly to the response prompt. Consequently, those conditions changed both the generator input and the number of document slots; their scores cannot isolate an effect of query reformulation.

2.5 Experimental Setup

Query Dataset

We evaluated a subset of 50 queries drawn from a recovered pool of 201 unique queries generated by the five biomedical researchers who later served as evaluators. Earlier project notes described the pool as 200 queries; the value 201 reported here follows the archived files (IDs 0–200). Queries include questions posed to large language models during ongoing research as well as anticipated future questions, spanning disease pathogenesis, biochemical pathways, cellular mechanisms, transcriptional regulation, diagnostic markers, experimental model comparisons, and therapeutic strategies. The procedure used to select the 50-query human-evaluation subset is not fully documented, so it should not be assumed to be a random or representative sample.

Knowledge Base

To construct the external document corpus, we collected PubMed article abstracts spanning 2015–2024 using the Biopython Entrez API. The corpus used abstracts rather than full texts; this experiment did not compare document lengths or corpus construction strategies. Literature retrieval used the search term "Alzheimer's disease". The abstracts were processed into text chunks prior to embedding. The corpus comprised approximately 18,000 PubMed abstracts after collection; exact deduplication procedures and chunking parameters are not fully recovered from available records and represent a reproducibility limitation.

Implementation Details

For retrieval, we employed BM25 [12] as a sparse retriever using default parameters (without specific tuning for k_1 or b) and DPR [13] as a dense retriever using the `bert-base-encoder.cp` checkpoint. For response generation, we used GPT-4o-mini as the generation model. For keyword generation and query transformation, GPT-4o-mini was used with temperature = 0 (with all other parameters left at default) and a fixed output format specified in the prompt. No post-hoc candidate ranking, semantic filtering, or hard truncation of excess keywords was applied. Duplicate handling was performed within each two-step stream; duplicates could remain across keyword streams. Retrieval scores and PubMed identifiers were not retained in the evaluation archive.

2.6 Evaluation Metrics

We evaluated the quality of generated responses using the following four dimensions, assessed on a five-point Likert scale (1 = very poor, 5 = excellent) by both human experts and an automated LLM judge.

- **Factuality**: Whether the response provides factually accurate information without introducing false statements with respect to the given query.
- **Diversity**: Whether the response reflects a broad range of information, perspectives, approaches, or aspects in addressing the given query.
- **Clarity**: Whether the response is clearly structured, logically organized, and easy to understand.
- **Insightfulness**: Whether the response goes beyond surface-level information and offers meaningful insights or perspectives that help interpret or resolve the query.

2.7 Human Evaluation Protocol

Five biomedical researchers associated with ongoing Alzheimer's disease research independently evaluated the 50-query subset. These same five researchers had generated the query pool. Each evaluator received a structured Excel file containing all eight reformulation conditions for each query, presented side by side to facilitate within-query comparison. For each condition, evaluators were shown the original query, the generated response, and the archived retrieved abstracts for that condition: two slots for no-keyword conditions, usually four for keyword conditions, and ten for the two five-keyword exceptions. Evaluators assigned integer scores (1–5) on each of the four dimensions using the rubric described above. The materials distributed to evaluators concealed the identity of the retrieval and reformulation condition that produced each response. Condition-order randomisation and formal rater training are not documented.

The evaluation was conducted after the automated GPT-4o evaluation had been completed; it was not pre-registered as the primary analysis. We therefore treat the human-evaluation analysis as exploratory. There were 2,000 query–condition–rater records and 8,000 possible scalar scores. A missing query-text cell in one evaluator file was linked by its query identifier after verifying that all eight response and context texts matched the other files; its available ratings were retained. Missing scores were excluded without imputation; missingness and complete-case sample sizes are reported in Table 3. Evaluator-level data are released only under stable pseudonymous identifiers.

2.8 Automated Evaluation

Automated evaluation was conducted using GPT-4o [14] as an LLM judge, assigning scores on the same five-point Likert scale under each metric. For each rating, GPT-4o received the original query and generated response but was not shown the retrieved abstracts. Responses were scored individually rather than in the side-by-side condition display used for expert evaluation. Thus, evaluator type was confounded with evidence access and presentation format. The judge prompt, rubric, and exact scoring instructions are shown in the Appendix. All automated results should be interpreted as descriptive comparisons between the two realised protocols, not as an isolated test of human-versus-LLM judgement.

2.9 Agreement Analysis

To characterise the relationship between the expert and automated scoring protocols, we computed Pearson and Spearman correlations between the available-rater mean and GPT score over the 400 query–condition pairs for each metric. Each pair receives equal weight. Human condition means in Table 2 instead pool available individual ratings, so their weighting differs slightly when scores are missing. Correlation is an association measure; mean human-minus-GPT differences describe scale offsets. These correlations should not be interpreted as isolating evaluator identity because the protocols differed in evidence access and presentation format. Condition-level ranking agreement was assessed via Spearman rank correlation across the eight conditions using full-precision condition means from Tables 2 and 4, with average ranks for ties. Inter-rater reliability among the five experts was assessed using the intraclass correlation coefficient (ICC(A,1): two-way random effects, absolute agreement, single rater) and Krippendorff's α (interval scale). Both reliability estimators used complete five-rater items, separately for each metric. Treating the ordinal scores as interval values is an analytic assumption; Spearman correlation provides a rank-based companion measure. Repeated queries across conditions and a fixed expert panel limit generalisation of these reliability point estimates. To account for repeated observation of each query across eight conditions, percentile 95% confidence intervals were calculated using a query-cluster bootstrap with 2,000 replicates (seed: 20260807); every resampled query retained all eight conditions. These intervals reflect query-sampling uncertainty after averaging experts, not uncertainty from sampling a different panel of evaluators. Analyses used the `pingouin`,

`krippendorff`, and `scipy` packages. Because the evaluation was post-hoc and exploratory, no multiplicity-adjusted confirmatory tests are presented.

As post-hoc sensitivity checks, we repeated the pooled correlations using only items with all five expert ratings, recomputed panel means after excluding each rater in turn, and calculated correlations separately within each condition. These checks assess dependence on missingness, panel composition, and pooling across configurations. They are descriptive analyses of the existing archive, without additional model calls or hypothesis tests.

3 Results

3.1 Response-Level Associations and System Rankings

Table 1 summarises the query–condition-level correlation between expert average scores and GPT scores across the eight shared conditions (50 query clusters; 400 query–condition pairs). Pearson r values ranged from 0.28 (factuality; query-cluster 95% CI: 0.13–0.43) to 0.54 (diversity; 0.43–0.64), and Spearman ρ values ranged from 0.27 to 0.52. These values indicate modest association across query–condition pairs; they do not measure absolute agreement.

Condition-level ranking agreement was higher: Spearman ρ across the eight conditions was 0.93 for clarity, 0.86 for insightfulness, 0.71 for diversity, and 0.67 for factuality. Thus, while GPT and human experts broadly agreed on the *relative ordering* of conditions by clarity and insightfulness, their ordering of conditions by factuality was less consistent.

Directional differences were also observed. Averaged across the shared conditions, the human-minus-GPT difference was +0.63 points for factuality (query-cluster 95% CI: 0.44–0.81), +0.13 for diversity (0.02–0.24), −0.52 for clarity (−0.63 to −0.40), and −0.52 for insightfulness (−0.61 to −0.44). The sign was consistent across the eight conditions for factuality, clarity, and insightfulness. These differences show systematic score offsets between the protocols; they do not establish which source was more accurate.

Table 1: Agreement and directional score differences between expert averages and GPT-4o. Confidence intervals are percentile intervals from 2,000 query-cluster bootstrap replicates; all eight conditions were retained within a resampled query. H–G denotes the mean human-minus-GPT score.

Metric	Pearson r [95% CI]	Spearman ρ [95% CI]	H–G mean [95% CI]
Factuality	0.28 [0.13, 0.43]	0.28 [0.13, 0.42]	0.63 [0.44, 0.81]
Diversity	0.54 [0.43, 0.64]	0.52 [0.40, 0.63]	0.13 [0.02, 0.24]
Clarity	0.30 [0.17, 0.42]	0.27 [0.14, 0.40]	-0.52 [-0.63, -0.40]
Insightfulness	0.39 [0.29, 0.48]	0.37 [0.28, 0.45]	-0.52 [-0.61, -0.44]

3.2 Sensitivity Checks

Complete-case Pearson correlations were 0.28 for factuality, 0.56 for diversity, 0.30 for clarity, and 0.39 for insightfulness, close to the primary estimates. Excluding one expert at a time yielded ranges of 0.16–0.31, 0.51–0.53, 0.21–0.33, and 0.35–0.40, respectively. Thus, the magnitude of the factuality and clarity associations depended partly on panel composition, while none of the omissions produced uniformly high correlations.

Within individual conditions (50 queries each), Pearson correlations ranged from 0.11 to 0.33 for factuality, 0.27 to 0.68 for diversity, 0.05 to 0.46 for clarity, and 0.21 to 0.53 for insightfulness. The pooled estimates therefore conceal variation across configurations. These ranges are observed point estimates, not confidence intervals; the full Pearson, Spearman, and score-offset results are provided in the ancillary files.

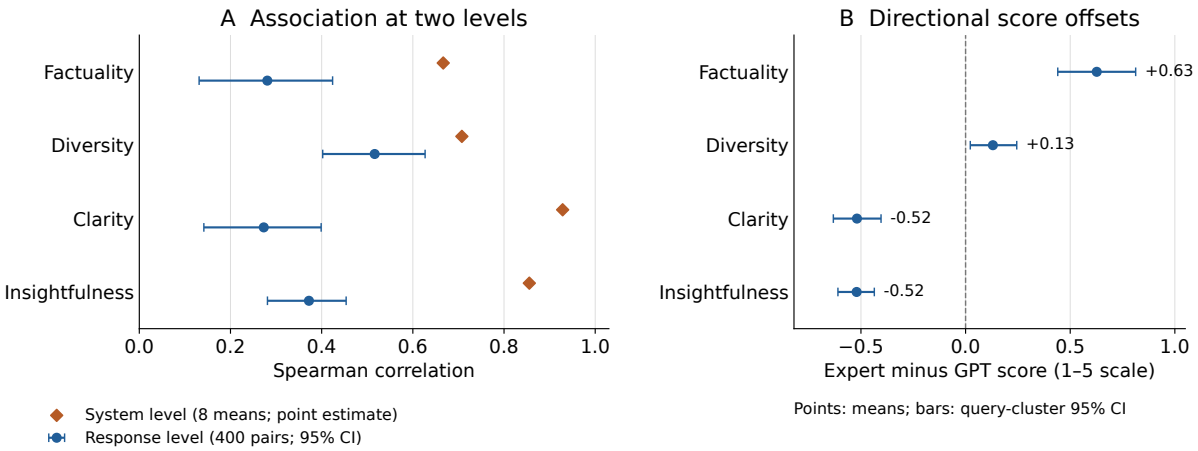


Figure 2: Two levels of score association and directional offsets. Left: Spearman correlations across 400 query–condition pairs (expert-panel means versus GPT-4o scores) and across eight configuration means. Only response-level estimates have query-cluster 95% intervals; configuration-level points are descriptive and use eight averages. The coefficients estimate different quantities, so their separation is not a significance test. Right: mean expert-minus-GPT score differences, with query-cluster 95% intervals; zero denotes no mean offset. Both panels use the same archived responses.

3.3 Human Evaluation Results

Table 2 reports descriptive average expert scores across the eight shared conditions. These are exploratory aggregate results; given low inter-rater agreement, they should not be interpreted as stable ground-truth estimates. Under BM25, keyword-augmented HyDE conditions tended to receive the highest expert scores on factuality ($\bar{x} = 4.62$) and diversity ($\bar{x} = 3.67$). Under DPR, differences across conditions were smaller in magnitude. Expert scores were notably higher on factuality (overall $\bar{x} = 4.42$) than on insightfulness ($\bar{x} = 3.38$). This difference in mean score should not be interpreted as greater consistency: factuality had the lowest single-rater ICC.

Inter-rater agreement was low for individual raters across all four metrics: ICC(A,1) ranged from 0.09 (factuality) to 0.24 (diversity), and Krippendorff's α ranged from 0.02 (factuality) to 0.21 (diversity). These values indicate that individual expert ratings varied substantially even for the same response, which substantially limits the interpretability of condition-level differences. Descriptive differences across conditions should be understood as reflecting this uncertainty.

3.4 Automated GPT Evaluation Results

Table 4 reports GPT-4o judge scores for the same 50-query subset and eight conditions. GPT scores on clarity were consistently higher than expert scores (GPT $\bar{x} = 4.02$–4.34 vs. expert $\bar{x} = 3.39$–4.04), while GPT scores on factuality were consistently lower than expert scores (GPT $\bar{x} = 3.64$–4.00 vs. expert $\bar{x} = 4.30$–4.62). Under BM25, MeSH-guided HyDE achieved the highest GPT score on diversity (3.64); under DPR, keyword HyDE showed the highest factuality (3.74) and clarity (4.24).

Table 2: Descriptive average expert scores (5 evaluators, 1–5 Likert scale) for the eight shared conditions. Values pool all available evaluator scores within each condition (up to 250 per metric); missing scores are excluded. These are exploratory observations; low inter-rater agreement (ICC(A,1): 0.09–0.24) limits interpretability.

Condition	Factuality	Diversity	Clarity	Insightfulness
Retriever: BM25				
HyDE (no keyword)	4.50	3.48	3.72	3.44
Naive (no keyword)	4.30	3.27	3.40	3.17
HyDE + Keyword	4.62	3.64	4.04	3.61
MeSH-guided + HyDE	4.53	3.67	4.02	3.53
Retriever: DPR				
HyDE (no keyword)	4.37	3.51	3.51	3.27
Naive (no keyword)	4.32	3.40	3.39	3.17
HyDE + Keyword	4.39	3.44	3.67	3.41
MeSH-guided + HyDE	4.35	3.64	3.66	3.42

Table 3: Missing scalar scores and expert inter-rater reliability. Each metric has 2,000 possible scores; reliability uses items with all five ratings. ICC(A,1) is single-rater absolute agreement; ICC(A,5) is agreement for the panel mean. Alpha uses interval distances. These are descriptive point estimates.

Metric	Missing	Items	ICC(A,1)	ICC(A,5)	α
Factuality	15	387	0.09	0.34	0.02
Diversity	15	387	0.24	0.61	0.21
Clarity	16	386	0.15	0.47	0.11
Insightfulness	15	387	0.11	0.38	0.07

Table 4: Descriptive average GPT-4o judge scores (1–5 Likert scale) for the eight shared conditions on the 50-query subset. Each condition includes 50 responses.

Condition	Factuality	Diversity	Clarity	Insightfulness
Retriever: BM25				
HyDE (no keyword)	3.86	3.42	4.22	4.00
Naive (no keyword)	3.78	3.14	4.02	3.76
HyDE + Keyword	3.98	3.56	4.32	4.00
MeSH-guided + HyDE	4.00	3.64	4.34	3.98
Retriever: DPR				
HyDE (no keyword)	3.64	3.14	4.14	3.76
Naive (no keyword)	3.66	3.14	4.10	3.84
HyDE + Keyword	3.74	3.50	4.24	3.96
MeSH-guided + HyDE	3.70	3.46	4.18	3.88

4 Discussion

System Rankings and Individual Responses Answer Different Questions

The main result is a distinction between two levels of evaluation. Across eight configurations, expert and GPT-4o mean scores gave broadly similar rankings, particularly for clarity and insightfulness. Across the individual query–condition pairs, associations were weaker. The coefficients describe different quantities, and their numerical contrast is not a test of a statistically significant difference. Nevertheless, reporting only configuration rankings would omit substantial variation in the correspondence of individual response scores.

This distinction matters when choosing how to use an automated evaluator. A comparison of system averages informs configuration selection; it does not by itself validate the evaluator's assessment of a particular response. In this archive, a report containing both levels of association, directional offsets, and expert reliability gives a more useful account than a single human–LLM correlation. Figure 2 summarises the two levels and the score offsets on a common set of responses.

Scoring Dimensions and Panel Composition

The expert panel assigned higher factuality scores than GPT-4o, whereas GPT-4o assigned higher clarity and insightfulness scores. Thus, the relationship between the protocols was dimension-specific rather than a uniform tendency for one source to score more generously. Diversity had the largest pooled response-level correlation and the smallest mean offset. These observations suggest examining each rubric dimension separately when validating an automated evaluation protocol.

The sensitivity checks help locate the uncertainty in this comparison. Complete-case results were close to the main estimates, so excluding the observed incomplete items did not materially change the associations. Rater omission affected factuality and clarity more than diversity and insightfulness. Together with single-rater ICC(A,1) of 0.09–0.24 and panel-mean ICC(A,5) of 0.34–0.61, this shows why the composition and reliability of the expert panel belong in the evaluation report. Differences in scale use, rubric interpretation, or restricted score variation may contribute, but the scores do not distinguish those explanations.

What the Configuration Comparisons Establish

Keyword-augmented HyDE configurations received descriptively higher scores under BM25, but they also supplied more document slots and explicit keywords to the generator. The BM25 no-keyword implementation had a second-step selection artefact. These factors prevent attribution of the score differences to query reformulation or retriever type. The archived outputs remain useful as a shared set of responses for comparing scoring protocols, but they do not identify an optimal RAG configuration.

Limitations

The study covers one domain, 50 queries selected by an undocumented procedure, and one expert panel whose members also authored the query pool. Familiarity with the questions may affect ratings. The retrospective analysis was not pre-registered, and condition-order randomisation and formal rater training are not documented. Generalisation to independently sampled queries and evaluators is therefore limited. The query-cluster intervals account for repeated conditions within queries but do not quantify uncertainty from recruiting a different expert panel.

GPT-4o and the generator, GPT-4o-mini, belong to the same model family. Same-family preference is a possible source of judging bias [9], but this study cannot determine its presence or direction. Only archived judge scores are available; repeated scoring and alternative judge

families were not evaluated. The non-equivalent evidence and presentation protocols also prevent a clean comparison of evaluator identity.

The scores are ordinal. Means, Pearson correlations, ICC, and interval alpha assume interval-scale treatment; Spearman correlations provide a rank-based companion analysis. No independent factuality labels or retrieval relevance gold standard was collected. Retrieval scores, ranked candidate lists, and PubMed identifiers were not retained, so retrieval effectiveness cannot be reconstructed from these outputs. Exact model snapshots and some corpus-processing settings are also missing. The public release reproduces the statistical analysis of ratings; it excludes query and response text and cannot support independent content adjudication or exact end-to-end pipeline reproduction.

A follow-up study should first match evidence access and presentation across judges, calibrate the expert rubric, and recruit an independent panel. Testing reformulation effects would additionally require a fixed document budget, separation of retrieval and generation-prompt changes, correction of the BM25 selection artefact, and retained relevance labels and retrieval rankings.

5 Conclusion

This study provides a reproducible expert–LLM evaluation case study for Alzheimer's disease literature RAG. The same eight configurations yielded relatively similar expert and GPT-4o system rankings, weaker associations on individual responses, and dimension-specific score offsets. Complete-case analyses preserved the main pattern, while rater-omission analyses identified sensitivity to panel composition. These findings support evaluating automated judges at both the system and response levels and reporting score offsets together with expert reliability. The released paired ratings make those comparisons independently reproducible within the scope of the two observed evaluation protocols.

Ethics and Consent

Five adult biomedical researchers participated as evaluators and consented to research use and public reporting of their queries and ratings. The evaluation involved no patients, clinical data, or other sensitive personal information. Evaluator-level data are released under pseudonymous identifiers.

Author Roles and Competing Interests

Two authors (K.S.C. and B.C.) were among the five evaluators. The authors declare no other competing interests.

Appendix A Prompts and Implementation Details

This appendix reproduces the prompts used in each pipeline component as screenshot images. All prompts were used with GPT-4o-mini (generation and reformulation) or GPT-4o (evaluation judge).

Recovered parameters: Temperature = 0 for keyword generation and query transformation, with all other sampling parameters left at defaults. The prompt requested up to two keywords, but the archived parser retained five for two queries. No post-hoc candidate ranking or hard truncation was applied. DPR used the `bert-base-encoder.cp` checkpoint. BM25 used default parameters (k_1, b).

A.1 Keyword Generation

Prompt: Generate General Keywords

Instruction: Generate a concise set of precise and domain-specific keywords up to {num} based on the given query. Focus only on the most critical and high-impact keywords that are essential for understanding or addressing the query.
Each keyword should represent the core elements of the query, such as key mechanisms, biological processes, or specific interactions. Exclude minor or overly generic terms like 'relationship' or 'mechanism' unless they are explicitly critical to the query. Prioritize terms that are unique, actionable, and directly relevant to the domain of the query.
Provide the results as a numbered list in the format: <number>. <keyword>.

Query: {query}

Figure A1: Prompt template for key-concept-based keyword generation (Part 1).

Prompt: Generate Mesh Keywords

Instruction: Generate a set of keywords up to {num} based on the given query, leveraging the MeSH (Medical Subject Headings) vocabulary. The keywords should be derived from relevant MeSH terms or closely related concepts. Reflect the core elements of the query, including biological processes, diseases, genes, proteins, and pathways. Avoid overly generic terms unless explicitly relevant to the query.
Prioritize specific MeSH terms that align directly with the query. Terms that are actionable and provide deeper insight into the topic.
Output the results as a numbered list in the format <number>. <MeSH term>.

Query: {query}

Figure A2: Prompt template for MeSH-guided keyword generation (Part 2).

A.2 Query Transformation

Prompt: Pseudo-Document Generation (HyDE)

Instruction: Please write down an abstract of a biology or medical paper that can solve the given query by using information or topic from abstract.

Query: {query}

Figure A3: Archived prompt for HyDE without keywords.

Prompt: Pseudo-Document Generation (HyDE) with Keyword

Instruction: Generate only one abstract of a biology or medical paper that can solve the given query based on the keyword. The abstract should directly address the query while focusing solely on the perspective of the keyword. Ensure that the abstract is logically structured, concise, and relevant to both the query and aspect. Provide the abstract text as plain text without titles, numbering, or additional formatting.

Query: {query}
Keyword: {keyword}

Figure A4: Archived prompt for keyword-conditioned HyDE.

Prompt: Query Rewriting

Instruction: Generate a refined and answerable question based on the given query. The revised question should eliminate ambiguity, focus on concrete and practical aspects, and ensure clarity in scope and intent. Avoid abstract or overly philosophical interpretations and instead reframe the query to facilitate a clear and actionable response.

Query: {query}

Figure A5: Archived prompt for query rewriting without keywords (not included in the expert comparison).

Prompt: Query Rewriting with Keyword

Instruction: Generate only one question based on the given query. The question should focus on the given keyword of the topic. Output only the question text as plain text, without any titles, numbering, or additional formatting.

Query: {query}
Keyword: {keyword}

Figure A6: Archived prompt for keyword-conditioned query rewriting (not included in the expert comparison).

A.3 Generation

Prompt: Generate without Keywords

Instruction: Using the provided abstracts, generate a response that either addresses the given query directly or provides meaningful insights that contribute to understanding or resolving the query.

Query: {query}
Abstracts: {abstracts}

Figure A7: Archived response-generation prompt without keywords.

Prompt: Generate with Keywords

Instruction: Each keyword corresponds to abstracts in one-to-one relationship. Using the provided keywords and their corresponding abstracts, generate a response that either addresses the given query directly or offers meaningful insights that contribute to understanding or resolving the query.

Query: {query}
Keywords: {keywords}
Abstracts: {abstracts}

Figure A8: Archived response-generation prompt with keywords.

A.4 Evaluation

Prompt: Evaluate Factuality

Instruction: Evaluate the generated response based on Factuality, which measures the extent to which the response provides accurate, truthful, and verifiable information without introducing false, misleading, or hallucinated content in relation to the given query.

Use the following 5-point scale to assign a Factuality score (1 to 5 points):

1 Point (Very Low Factuality)
- The response contains multiple factual errors, hallucinations, or clearly incorrect statements.
- It misrepresents key facts and provides false or misleading information that may significantly harm understanding or decision-making.

2 Points (Low Factuality)
- The response has some accurate content, but also includes important factual inaccuracies, unsupported claims, or misleading generalizations.
- These issues reduce the reliability of the response.

3 Points (Moderate Factuality)
- The response is mostly accurate, but may contain minor factual inaccuracies, vague statements, or claims lacking clear evidence.
- While not seriously misleading, it requires some fact-checking or clarification.

4 Points (High Factuality)
- The response is factually accurate overall, with reliable and verifiable content.
- There may be minor omissions or simplifications, but no false or misleading claims.

5 Points (Very High Factuality)
- The response is entirely accurate, well-supported by scientific evidence, and free from errors, hallucinations, or exaggeration.
- All claims are trustworthy and align with established biomedical knowledge.

Evaluation Task:
After reviewing the response, assign a Factuality score (1 to 5 points) based on the scale above. Only provide the score. No justification is required.

Query: `{query}`
Response: `{response}`

Figure A9: GPT-4o evaluation judge prompt for the factuality metric.

Prompt: Evaluate Diversity

Instruction: Evaluate the generated response based on Diversity, which measures the extent to which the response integrates a broad range of relevant information, perspectives, biological mechanisms, diagnostic approaches, methodologies, or treatment strategies to effectively address the given query.

Use the following 5-point scale to assign a Diversity score (1 to 5):

1 Point (Very Low Diversity)
- The response relies heavily on a single concept, source, or perspective, with little to no exploration of other relevant aspects or alternative viewpoints.
- Critical dimensions relevant to the query, such as underlying mechanisms, diagnostic considerations, or therapeutic options, are largely absent.

2 Points (Low Diversity)
- The response mentions more than one piece of information, but coverage is narrow, repetitive, or focused on a single aspect.
- Other important aspects (e.g., additional biological pathways, diagnostic tools, or treatment approaches) are underdeveloped or missing.

3 Points (Moderate Diversity)
- The response addresses multiple relevant aspects related to the query, such as mechanisms, diagnostic approaches, or treatment strategies, but lacks depth or balance in some areas.
- There is some variety, but the range of perspectives or approaches could be expanded to provide a more comprehensive understanding.

4 Points (High Diversity)
- The response incorporates a variety of relevant biological, diagnostic, and/or therapeutic perspectives, covering most major aspects related to the query.
- The response is well-rounded, with only minor gaps or slight imbalances in coverage.

5 Points (Very High Diversity)
- The response synthesizes a comprehensive range of biological, diagnostic, and therapeutic aspects, drawing from multiple perspectives and approaches.
- All critical dimensions relevant to the query are addressed, and the information is balanced and mutually complementary.

Evaluation Task:
After reviewing the response, assign a Diversity score (1 to 5 points) based on the scale above.
Only provide the score. No justification is required.

Query: `{query}`
Response: `{response}`

Figure A10: GPT-4o evaluation judge prompt for the diversity metric.

Prompt: Evaluate Clarity

Instruction: Evaluate the generated response based on Clarity, which measures the extent to which the response is expressed in a clear, well-structured, and easy-to-understand manner, helping the reader grasp the solution to the given query without confusion.

Use the following 5-point scale to assign a Clarity score (1 to 5):

1 Point (Very Low Clarity)
- The response is highly difficult to understand, disorganized, or contains excessive jargon and unclear explanations.
- The core message is obscured, making it hard to grasp how it addresses the query.

2 Points (Low Clarity)
- The response is partially understandable, but contains confusing sections, poor structure, or vague language.
- Significant effort is required to comprehend how the response relates to the query.

3 Points (Moderate Clarity)
- The response is generally understandable, but some sentences may be difficult to follow, or the structure could be improved for easier reading.
- Occasional ambiguity or complex phrasing may slightly hinder understanding.

4 Points (High Clarity)
- The response is mostly clear and easy to understand, with logical structure and minimal ambiguity.
- Minor refinements could further enhance readability, but comprehension is smooth overall.

5 Points (Very High Clarity)
- The response is exceptionally clear, concise, and well-structured, with straightforward language and smooth logical flow.
- The reader can easily understand how the response addresses the query without any confusion.

Evaluation Task:
After reviewing the response, assign a Clarity score (1 to 5 points) based on the scale above.
Only provide the score. No justification is required.

Query: `{query}`
Response: `{response}`

Figure A11: GPT-4o evaluation judge prompt for the clarity metric.

Prompt: Evaluate Insightfulness

Instruction: Evaluate the generated response based on Insightfulness, which measures the extent to which the response offers meaningful, valuable, and non-obvious insights that contribute to understanding or addressing the given query.

Use the following 5-point scale to assign an Insightfulness score (1 to 5 points):

1 Point (Very Low Insightfulness)
- The response provides little to no new understanding, stating only obvious or generic information.
- It fails to offer any useful guidance in addressing the query.

2 Points (Low Insightfulness)
- The response contains some relevant information, but the insights are shallow or largely restate well-known facts.
- There is minimal value added to deepen understanding of the query.

3 Points (Moderate Insightfulness)
- The response offers a few helpful insights, but they may be somewhat basic, incomplete, or lacking originality.
- Some value is added, but the response could be more thought-provoking or detailed.

4 Points (High Insightfulness)
- The response presents several meaningful insights, going beyond surface-level information.
- It enhances understanding of the query, though a few aspects might still benefit from deeper exploration.

5 Points (Very High Insightfulness)
- The response delivers deep, valuable, and non-obvious insights, offering novel perspectives or synthesizing information in a way that greatly enhances understanding of the query.
- The response stands out as particularly insightful and well thought out.

Evaluation Task:
After reviewing the response, assign an Insightfulness score (1 to 5 points) based on the scale above. Only provide the score. No justification is required.

Query: `{query}`
Response: `{response}`

Figure A12: GPT-4o evaluation judge prompt for the insightfulness metric.

Appendix B Data and Code Availability

The manuscript, analysis scripts, and text-free rating data are available at `https://github.com/domino721/alzheimer-rag-evaluation`. The arXiv ancillary directory contains pseudonymised evaluator-level scores (2,000 records), a text-free table of expert averages and GPT scores (400 query–condition pairs), and executable scripts that reproduce the descriptive means, inter-rater reliability, rank associations, query-cluster bootstrap, and sensitivity checks. Stable query and rater identifiers support linkage without including evaluator names, query text, generated responses, or retrieved abstract text. These materials make the statistical reanalysis self-contained; they do not reproduce the original model calls.

The legacy pipeline is archived at `https://github.com/yeomsee/best_practices_bio_rag`, commit `25be531c6e8b5aff76b1a99a9011665e062299e2`. The corpus and DPR index are hosted at `https://huggingface.co/datasets/stv10121/pubmed_alzheimer_kb`. Exact model snapshots, execution dates, corpus deduplication and chunking parameters, and the final post-processing document count remain incompletely documented. No claim of exact end-to-end reproduction is made.

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